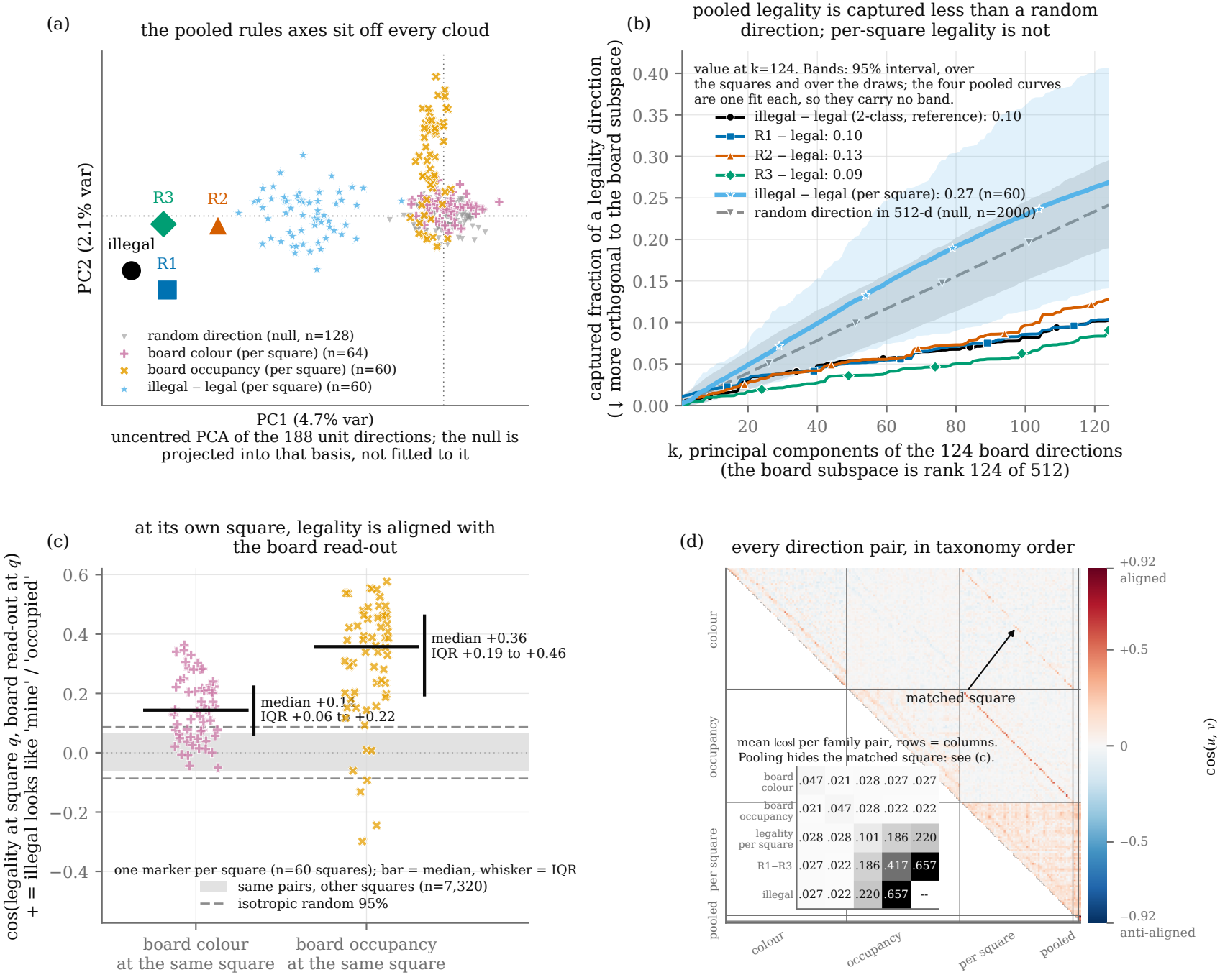


board colour — $\ell_{\text{mine}} - \ell_{\text{theirs}}$ at one square: which player's disc sits there.

board occupancy — $\frac{1}{2}(\ell_{\text{mine}} + \ell_{\text{theirs}}) - \ell_{\text{empty}}$ at one square: whether a disc sits there.

per-square legality — illegal – legal, refit on the rows where that square was the candidate move; the committed rules probes pool over all squares.

R1 placement — a move puts one disc of the mover's color on an empty square.

R2 bracketing — the disc must close a gapless line of opponent discs with one of the mover's own.

R3 turn — colors alternate; legality is judged for the mover, not the opponent.